



OpenFOAM 해석 환경 구축 및 DTLreactingFoam 설치 가이드

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

WSL 설치 가이드: Windows 기반 Linux 환경 구축

WSL 설치 가이드

WSL (Windows Subsystem for Linux)

- ✓ Windows 환경에서 별도의 듀얼 부팅이나 가상 머신 없이 Linux를 사용할 수 있도록 Microsoft에서 제공하는 기능임
- ✓ Ubuntu와 같은 Linux 배포판을 설치하여 Linux 명령어, 개발 도구 및 OpenFOAM과 같은 Linux 기반 프로그램을 사용할 수 있음
- ✓ Windows 10 버전 2004 이상(빌드 19041 이상) 또는 Windows 11에서 지원됨



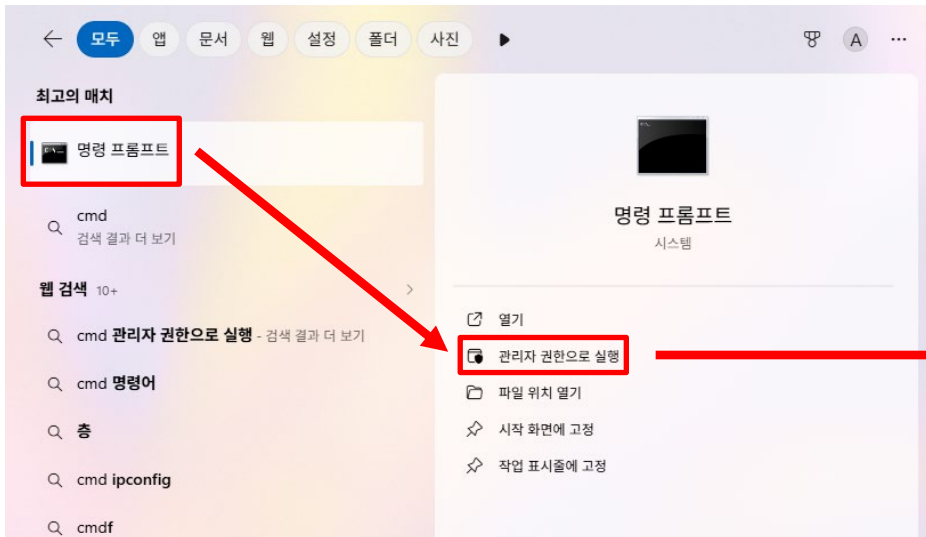
WSL 설치 가이드

Step 1. Windows Command Prompt (cmd) 실행

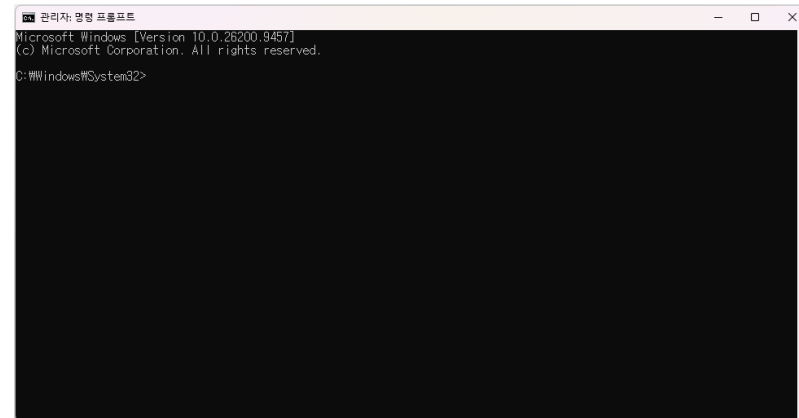
- ✓ Windows 탐색기 주소창에 'cmd' 입력



- ✓ 관리자 권한으로 Command Prompt 실행



Command Prompt



WSL 설치 가이드

Step 2. WSL (Windows Subsystem for Linux) 설치

- ✓ cmd에 아래 명령어를 입력하여 WSL과 Ubuntu 22.04 설치

```
ws1 --install -d Ubuntu-22.04
```

재시작 요청 시, Windows 재시작 후 명령어 입력

```
ws1 -d Ubuntu-22.04
```

아래 오류시 -d 지우고 -d로 변경

```
C:\Users\ghkrk>ws1 --install -d Ubuntu-22.04
잘못된 명령줄 인수: Ubuntu-22.04
지원되는 인수 목록을 가져오려면 'ws1.exe --help' 사용
하세요.
오류 코드: Wsl/E_INVALIDARG
이 오류가 예기치 않은 오류인 경우 기존 문제를 검색하
거나 https://github.com/microsoft/WSL/issues에서 새
문제를 제출하는 것이 좋습니다.
```

- ✓ Ubuntu 계정 이름 및 비밀번호 설정

```
Create a default Unix user account: mhs
```

 계정 이름 설정

```
New password:
Retype new password:
```

 비밀번호 설정

- ✓ WSL 및 Ubuntu 22.04 설치 완료 시 아래와 같이 표시됨

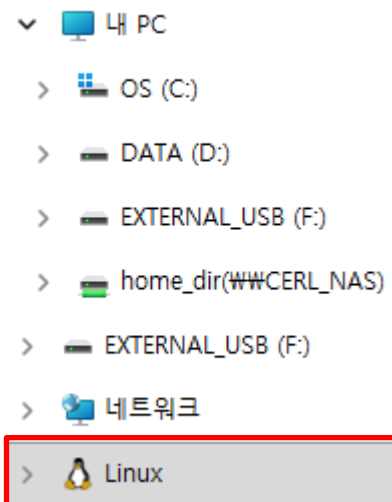
```
C:\ 선택 mhs@20245134-p01: /mnt/c/Windows/System32
mhs@20245134-p01: /mnt/c/Windows/System32$
```

이후부터는 cmd 실행 후 `ws1` 을 입력하여
WSL (Ubuntu) 환경을 실행가능

WSL 설치 가이드

Step 2. WSL (Windows Subsystem for Linux) 설치

- ✓ 설치 완료 후, 아래 그림과 같이 WSL의 Linux 폴더가 생성됨



- ✓ Linux 폴더를 통해 WSL에서 작업한 파일을 Windows로 쉽게 옮길 수 있음

OpenFOAM 설치 가이드

OpenFOAM 설치 가이드

OpenFOAM (OF) toolbox:

- OpenFOAM: **Open** Field **O**peration **A**nd **M**anipulation
- GNU General Public License(GPL)로 배포되는 오픈소스 패키지(C++)
- 셀 중심 유한체적법 cell centered Finite Volume Method 기반
- 화학반응, 난류, 열전달이 포함된 복잡한 유동을 포함한 다양한 물리 현상을 해석할 수 있는 폭넓은 기능 제공
- GNU GPL v3 라이선스에 따라 **무료로 다운로드**, 설치, 사용, 수정 및 배포 가능

OpenFOAM forks:



(official standard release)

Released Version*: 13 | **12** | 11 | 10 | 9 | 8 |
7 | 6 | 5.0 | 4.1 | 4.0 | 3.0.1 | 3.0.0 | 2.4.0 |
2.3.1 | 2.3.0 | 2.2.2 | 2.2.1 | 2.2.0 | 2.1.1 |
2.1.0 | 2.0.1 | 2.0.0 | 1.7.1 | 1.7.0 | 1.6 |
1.5 | 1.4.1 | 1.3 | 1.2 | 1.1 | 1.0

<https://openfoam.org>



(official development release)

Released Version: v2306 | v2212 | v2206 |
v2112 | v2106 | v2012 | v2006 | v1912 |
v1906 | v1812 | v1806 | v1712 | v1706 |
v1612+ | v1606+ | v3.0+

<https://www.openfoam.com>

foam-extend

(community driven
development branch)

Released Version:
4.1 | 4.0 | 3.2 | 3.1 | 3.0

<https://sourceforge.net/projects/foam-extend>

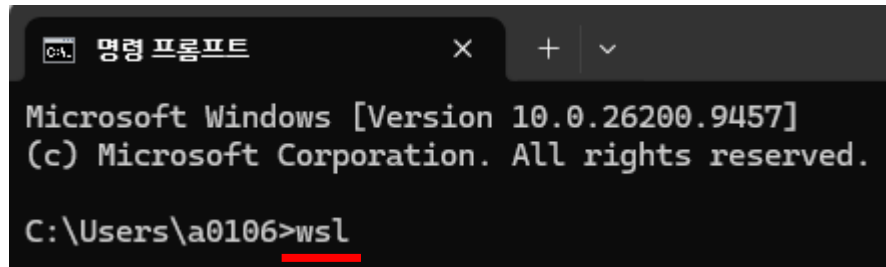
- **A library** for handling **surface chemistry** to simulate catalytic processes is not available in the official OpenFOAM.

OpenFOAM 설치 가이드

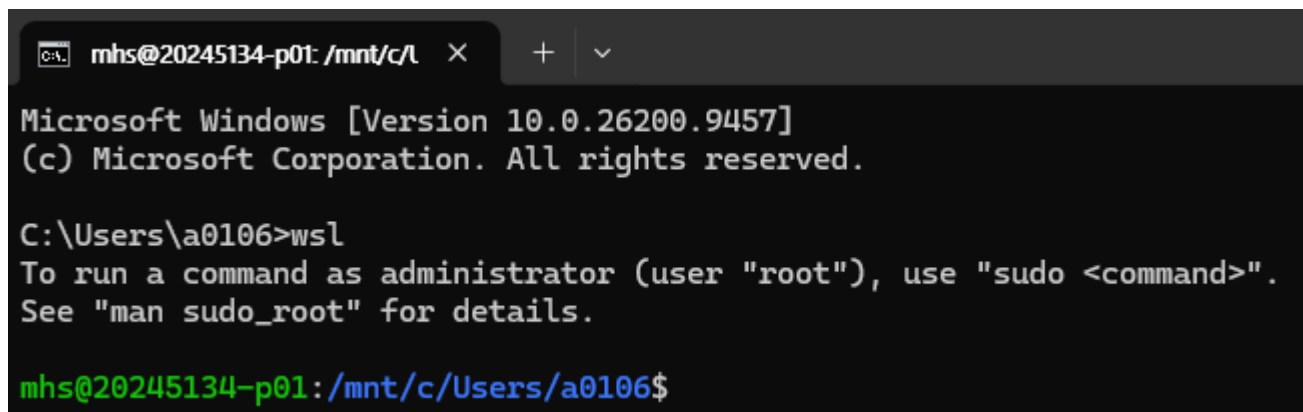
Step 1. WSL 실행

- ✓ cmd에서 아래 명령어로 WSL 실행

```
wsl
```



```
C:\Users\A0106>wsl
```



```
C:\Users\A0106>wsl
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\A0106>wsl
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

mhs@20245134-p01:/mnt/c/Users/A0106$
```

OpenFOAM 설치 가이드

Step 2. OpenFOAM 설치

✓ 아래 명령어를 순서대로 복사·붙여넣기하여 실행

비밀번호 요청 시, WSL 설치 과정에서 설정한 비밀번호 입력

1

```
sudo sh -c "wget -O - https://dl.openfoam.org/gpg.key > /etc/apt/trusted.gpg.d/openfoam.asc"
```

```
mhs@26245134-p01:/mnt/c/Users/a0186$ sudo sh -c "wget -O - https://dl.openfoam.org/gpg.key > /etc/apt/trusted.gpg.d/openfoam.asc"
--2026-09-21 19:11:46-- https://dl.openfoam.org/gpg.key
Resolving dl.openfoam.org (dl.openfoam.org)... 13.41.234.222
Connecting to dl.openfoam.org (dl.openfoam.org)|13.41.234.222|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1710 (1.7K) [application/pgp-keys]
Saving to: 'STDOUT'

-
100%[=====] 1.67K --.-KB/s in 0s

2026-09-21 19:11:47 (371 MB/s) - written to stdout [1710/1710]
```

정상 실행 완료 화면

2

```
sudo add-apt-repository http://dl.openfoam.org/ubuntu
```

```
mhs@26245134-p01:/mnt/c/Users/a0186$ sudo add-apt-repository http://dl.openfoam.org/ubuntu
Repository: 'deb http://dl.openfoam.org/ubuntu jammy main'
Description:
Archive for codename: jammy components: main
More info: http://dl.openfoam.org/ubuntu
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
Adding deb entry to /etc/apt/sources.list.d/archive_uri-http_d1_openfoam_org_ubuntu-jammy.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/archive_uri-http_d1_openfoam_org_ubuntu-jammy.list
Get:1 http://dl.openfoam.org/ubuntu jammy InRelease [9045 B]
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:4 http://dl.openfoam.org/ubuntu jammy/main amd64 Packages [15.6 kB]
Hit:5 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Fetched 24.7 kB in 1s (18.4 kB/s)
Reading package lists... Done
```

정상 실행 완료 화면

OpenFOAM 설치 가이드

Step 2. OpenFOAM 설치

✓ 아래 명령어를 순서대로 복사·붙여넣기하여 실행

3

```
sudo apt update
```

```
mhs@20245134-p01:/mnt/c/Users/a0106$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://dl.openfoam.org/ubuntu jammy InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
143 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

정상 실행 완료 화면

설치에 시간이 소요될 수 있음

4

```
sudo apt -y install openfoam12
```

```
Setting up python3-pil.imagetk:amd64 (9.0.1-1ubuntu0.5) ...
Setting up python3-ufolib2 (0.13.1+dfsg1-1) ...
Setting up python3-fonttools (4.29.1-2build1) ...
Setting up python3-pil:amd64 (9.0.1-1ubuntu0.5) ...
Setting up python3-matplotlib (3.5.1-2build1) ...
Setting up paraview (5.10.0-rc1-1build3) ...
Setting up python3-paraview (5.10.0-rc1-1build3) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for udev (249.11-0ubuntu3.17) ...
Processing triggers for install-info (6.8-4build1) ...
mhs@20245134-p01:/mnt/c/Users/a0106$
```

정상 실행 완료 화면

OpenFOAM 설치 가이드

Step 2. OpenFOAM 설치

✓ 아래 명령어를 순서대로 복사·붙여넣기하여 실행

5 `echo ". /opt/openfoam12/etc/bashrc" >> ~/.bashrc`

6 `source ~/.bashrc`

7 `foamRun -help`

```
mhs@20245134-p01:~$ foamRun -help

Usage: foamRun [OPTIONS]
options:
  -case <dir>          specify alternate case directory, default is the cwd
  -fileHandler <handler>
                        override the fileHandler
  -hostRoots <((host1 dir1) .. (hostN dirN))>
                        slave root directories (per host) for distributed running
  -libs '("lib1.so" ... "libN.so")'
                        pre-load libraries
  -noFunctionObjects    do not execute functionObjects
  -parallel            run in parallel
  -roots <(dir1 .. dirN)>
                        slave root directories for distributed running
  -solver <name>       Solver name
  -srcDoc              display source code in browser
  -doc                 display application documentation in browser
  -help               print the usage

Using: OpenFOAM-12 (see https://openfoam.org)
Build: 12-86e126a7bc4d
```

OpenFOAM 12가 오류 없이 정상적으로 설치된 경우, `foamRun -help` 입력 시 왼쪽과 같은 화면이 출력됨

DTLreactingFoam 설치 가이드

DTLreactingFoam 설치 가이드

DTLreactingFoam

- ✓ OpenFOAM에서 층류 반응성 유동의 수송 현상을 정밀하게 모사하기 위해 detailed transport model(DTM)을 적용하고, FTM 및 coTHERM 기법을 통해 해석 정확도를 유지하면서 계산 비용을 절감한 custom framework

- ✓ Github

<https://github.com/CCERL/DTLreactingFoam>

- ✓ Citation

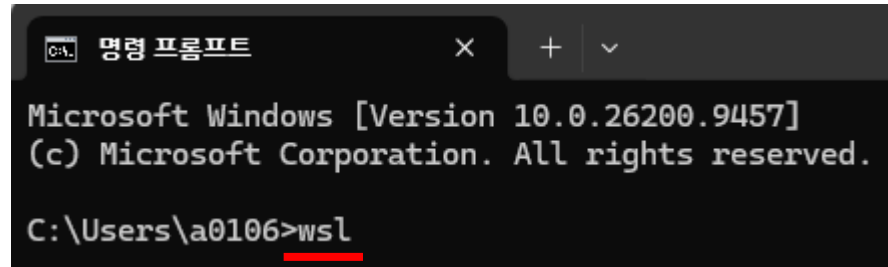
D. N. Nguyen, J. H. Lee, C. S. Yoo, DTLreactingFoam: An efficient CFD tool for laminar reacting flow simulations using detailed chemistry and transport with time-correlated thermophysical properties, *Computer Physics Communications* 322 (2026) 110052

DTLreactingFoam 설치 가이드

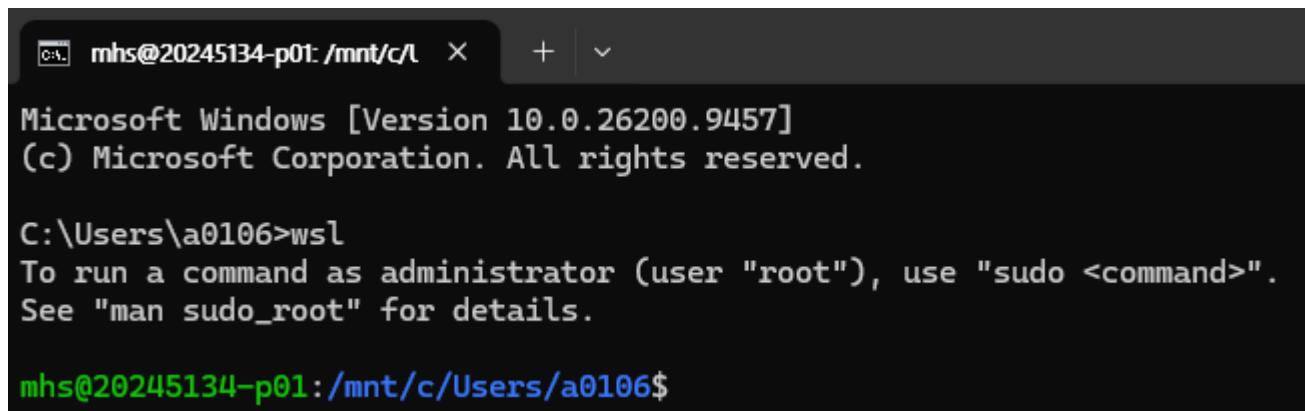
Step 1. WSL 실행

- ✓ cmd에서 아래 명령어로 WSL 실행

```
wsl
```



```
C:\Users\A0106>wsl
```



```
mhs@20245134-p01: /mnt/c/l X + v  
Microsoft Windows [Version 10.0.26200.9457]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\A0106>wsl  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
mhs@20245134-p01: /mnt/c/Users/a0106$
```

DTLreactingFoam 설치 가이드

Step 2. DTLreactingFoam 설치

✓ 아래 명령어를 순서대로 복사·붙여넣기하여 실행

1

```
cd
```

```
mhs@20245134-p01:/mnt/c/Users/a0106$ cd  
mhs@20245134-p01:~$
```

~는 Home directory를 의미

2

```
mkdir -p $FOAM_RUN
```

예시) 계정 이름이 mhs인 경우 mhs-12 디렉토리가 생성됨

```
mhs@20245134-p01:~$ cd $WM_PROJECT_USER_DIR  
mhs@20245134-p01:~/OpenFOAM/mhs-12$
```

DTLreactingFoam 설치 가이드

Step 2. DTLreactingFoam 설치

✓ 아래 명령어를 순서대로 복사·붙여넣기하여 실행

로컬 디렉토리 (ex: mhs-12) 에 DTLreactingFoam 프레임워크 설치

3

```
cd $WM_PROJECT_USER_DIR
```

```
mhs@20245134-p01:~$ cd $WM_PROJECT_USER_DIR
mhs@20245134-p01:~/OpenFOAM/mhs-12$ |
```

4

```
git clone https://github.com/danhnam11/DTLreactingFoam-12.git
```

```
mhs@20245134-p01:~/OpenFOAM/mhs-12$ git clone https://github.com/danhnam11/DTLreactingFoam-12.git
Cloning into 'DTLreactingFoam-12'...
remote: Enumerating objects: 1549, done.
remote: Counting objects: 100% (1549/1549), done.
remote: Compressing objects: 100% (796/796), done.
remote: Total 1549 (delta 755), reused 1527 (delta 737), pack-reused 0 (from 0)
Receiving objects: 100% (1549/1549), 1.67 MiB | 6.17 MiB/s, done.
Resolving deltas: 100% (755/755), done.
```

5

```
echo "export LIB_DTL12_SRC=$WM_PROJECT_USER_DIR/DTLreactingFoam-12/src/" >> ~/.bashrc
```

6

```
source ~/.bashrc
```

DTLreactingFoam 설치 가이드

Step 2. DTLreactingFoam 설치

✓ 아래 명령어를 순서대로 복사·붙여넣기하여 실행

7

```
cd $WM_PROJECT_USER_DIR/DTLreactingFoam-12/
```

```
mhs@20245134-p01:~/OpenFOAM/mhs-12$ cd $WM_PROJECT_USER_DIR/DTLreactingFoam-12/  
mhs@20245134-p01:~/OpenFOAM/mhs-12/DTLreactingFoam-12$ |
```

해당작업은 긴 시간이 소요됨

8

```
./Allwmake -j
```

DTLreactingFoam 설치 가이드

Step 3. DTLreactingFoam 설치 확인

✓ 정상적으로 완료되면, 라이브러리 파일들이 '\$FOAM_USER_LIBBIN'에 생성됨

```
cd $FOAM_USER_LIBBIN
```

```
mhs@20245134-p01:~/OpenFOAM/mhs-12/DTLreactingFoam-12$ cd $FOAM_USER_LIBBIN
mhs@20245134-p01:~/OpenFOAM/mhs-12/platforms/linux64GccDPInt32Opt/lib$ ll
total 135196
drwxr-xr-x 2 mhs mhs      4096 Sep 21 20:29 ./
drwxr-xr-x 4 mhs mhs      4096 Sep 21 20:29 ../
-rwxr-xr-x 1 mhs mhs    961632 Sep 21 20:29 libDTLreactingFoam.so*
-rwxr-xr-x 1 mhs mhs 19814248 Sep 21 20:21 libchemistryModel.so*
-rwxr-xr-x 1 mhs mhs 1850872 Sep 21 20:25 libcombustionModels.so*
-rwxr-xr-x 1 mhs mhs  386952 Sep 21 20:23 libcoupledThermophysicalTransportModels.so*
-rwxr-xr-x 1 mhs mhs 1099544 Sep 21 20:22 libfluidMulticomponentThermophysicalTransportModels.so*
-rwxr-xr-x 1 mhs mhs  738632 Sep 21 20:22 libfluidThermoThermophysicalTransportModels.so*
-rwxr-xr-x 1 mhs mhs 16560800 Sep 21 20:16 libfluidThermophysicalModels.so*
-rwxr-xr-x 1 mhs mhs  677568 Sep 21 20:22 libfluidThermophysicalTransportModel.so*
-rwxr-xr-x 1 mhs mhs  2366368 Sep 21 20:25 libfvConstraints.so*
-rwxr-xr-x 1 mhs mhs  4424680 Sep 21 20:27 libfvModels.so*
-rwxr-xr-x 1 mhs mhs  1213216 Sep 21 20:28 libinterRegionFvModels.so*
-rwxr-xr-x 1 mhs mhs  646392 Sep 21 20:28 libmulticomponentFluid.so*
-rwxr-xr-x 1 mhs mhs 72679624 Sep 21 20:19 libmulticomponentThermophysicalModels.so*
-rwxr-xr-x 1 mhs mhs  1118264 Sep 21 20:22 libphaseFluidMulticomponentThermophysicalTransportModels.so*
-rwxr-xr-x 1 mhs mhs  751000 Sep 21 20:22 libphaseFluidThermophysicalTransportModels.so*
-rwxr-xr-x 1 mhs mhs 1075280 Sep 21 20:23 libphaseSolidThermophysicalTransportModels.so*
-rwxr-xr-x 1 mhs mhs 1986952 Sep 21 20:24 libradiationModels.so*
-rwxr-xr-x 1 mhs mhs  527328 Sep 21 20:28 librotorDisk.so*
-rwxr-xr-x 1 mhs mhs  6103648 Sep 21 20:22 libsolidThermo.so*
-rwxr-xr-x 1 mhs mhs 1065312 Sep 21 20:22 libsolidThermophysicalTransportModels.so*
-rwxr-xr-x 1 mhs mhs  346032 Sep 21 20:14 libspecie.so*
-rwxr-xr-x 1 mhs mhs  473616 Sep 21 20:28 libspecieTransfer.so*
-rwxr-xr-x 1 mhs mhs 1474816 Sep 21 20:15 libthermophysicalProperties.so*
-rwxr-xr-x 1 mhs mhs  28752 Sep 21 20:22 libthermophysicalTransportModel.so*
```

DTLreactingFoam 설치 가이드

Step 3. DTLreactingFoam 설치 확인

✓ 정상적으로 완료되면, **실행 파일들**이 '\$FOAM_USER_APPBIN'에 생성됨

```
cd $FOAM_USER_APPBIN
```

```
mhs@20245134-p01:~/OpenFOAM/mhs-12/platforms/linux64GccDPInt320pt/bin$ cd $FOAM_USER_APPBIN
mhs@20245134-p01:~/OpenFOAM/mhs-12/platforms/linux64GccDPInt320pt/bin$ ll
total 2748
drwxr-xr-x 2 mhs mhs    4096 Sep 21 20:29 ./
drwxr-xr-x 4 mhs mhs    4096 Sep 21 20:29 ../
-rwxr-xr-x 1 mhs mhs 1260352 Sep 21 20:29 DTMchemkinToFoam*
-rwxr-xr-x 1 mhs mhs 295608 Sep 21 20:29 FTMchemkinToFoam*
-rwxr-xr-x 1 mhs mhs 1243256 Sep 21 20:29 chemkinToFoam*
```

ParaView 설치 가이드

ParaView 설치 가이드

ParaView

- ✓ 과학·공학 데이터의 시각화 및 후처리를 위한 오픈소스 프로그램
- ✓ OpenFOAM 등에서 계산 결과를 불러와 유동장, 압력, 속도, 화학종 분포 등을 시각적으로 분석할 수 있음
- ✓ 3D 결과 분석, slice, streamline, iso-surface 등 다양한 후처리 기능 제공



ParaView

ParaView 설치 가이드

Step 1. ParaView 설치 파일 다운로드

✓ 'ParaView-6.2.0-RC2-Windows-Python3.12-msvc2017-AMD64.msi' 다운로드

ParaView 설치 링크

<https://www.paraview.org/download/>

Get the Software

You can either download binaries or source code archives for the latest stable or previous release or access the current development (aka nightly) distribution through Git. Specific license information can be found [here](#). This software may not be exported in violation of any U.S. export laws or regulations. For more information regarding Export Control matters please go to https://kitware.com/export_control/index.html.

Version v6.2

Release Candidates

Sources **Windows** Linux macOS Silicon macOS Intel

Previews of ParaView's next release. Full suite of ParaView tools, including the ParaView GUI client, pypython, pvserver, and pvbatch. Versions with MPI in the name require [MS-MPI](#).

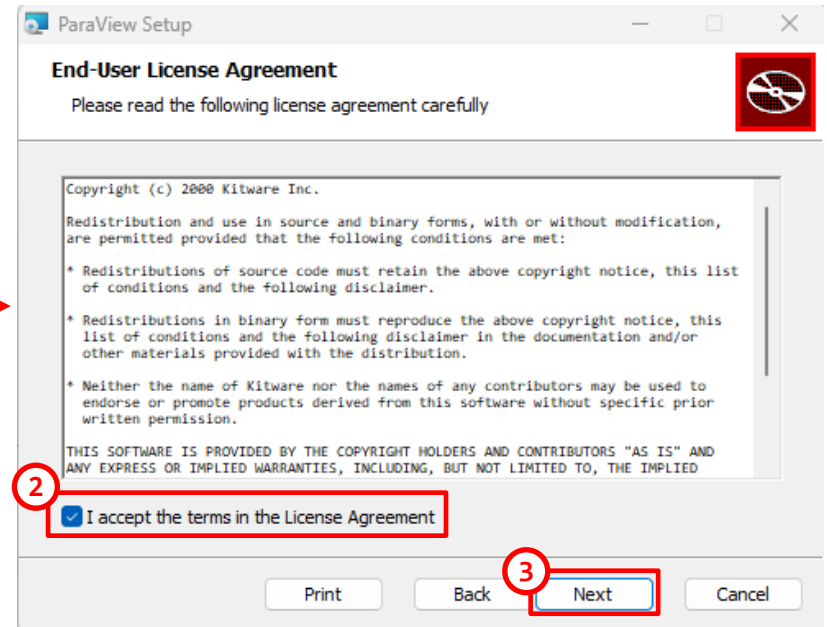
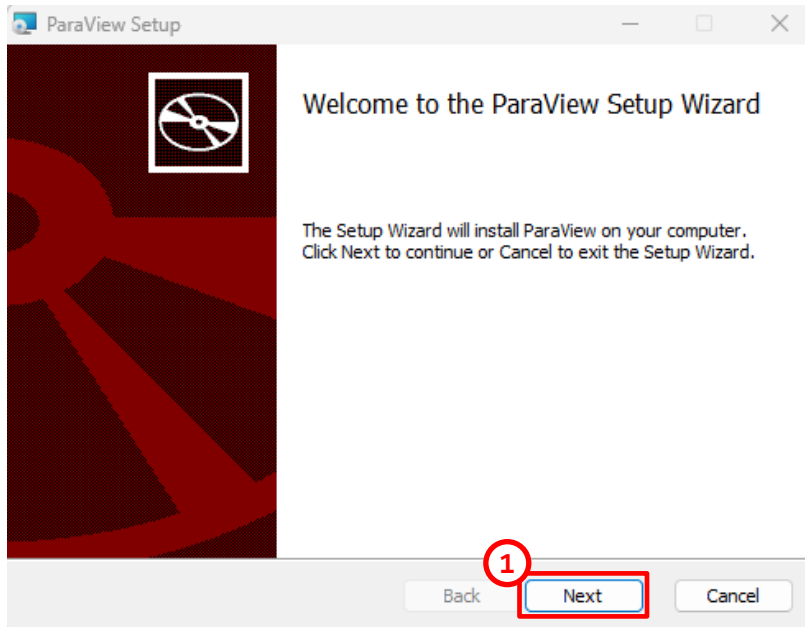
Unzip .zip packages in a directory with a very short path if you encounter an error message about a path being too long for some files.

📄 ParaView-6.2.0-RC2-Windows-Python3.12-msvc2017-AMD64.msi	2026-09-11 11:47	559.0M
📄 ParaView-6.2.0-RC2-MPI-Windows-Python3.12-msvc2017-AMD64.msi	2026-09-11 11:47	561.6M
📄 ParaView-6.2.0-RC2-Windows-Python3.12-msvc2017-AMD64.zip	2026-09-10 22:06	1.4G
📄 ParaView-6.2.0-RC2-MPI-Windows-Python3.12-msvc2017-AMD64.zip	2026-09-10 22:06	1.4G
📄 ParaView-6.2.0-RC1-Windows-Python3.12-msvc2017-AMD64.msi	2026-07-28 06:45	559.0M
📄 ParaView-6.2.0-RC1-MPI-Windows-Python3.12-msvc2017-AMD64.msi	2026-07-28 06:44	561.5M
📄 ParaView-6.2.0-RC1-Windows-Python3.12-msvc2017-AMD64.zip	2026-07-28 06:29	1.4G
📄 ParaView-6.2.0-RC1-MPI-Windows-Python3.12-msvc2017-AMD64.zip	2026-07-28 06:26	1.4G

ParaView 설치 가이드

Step 2. ParaView 설치

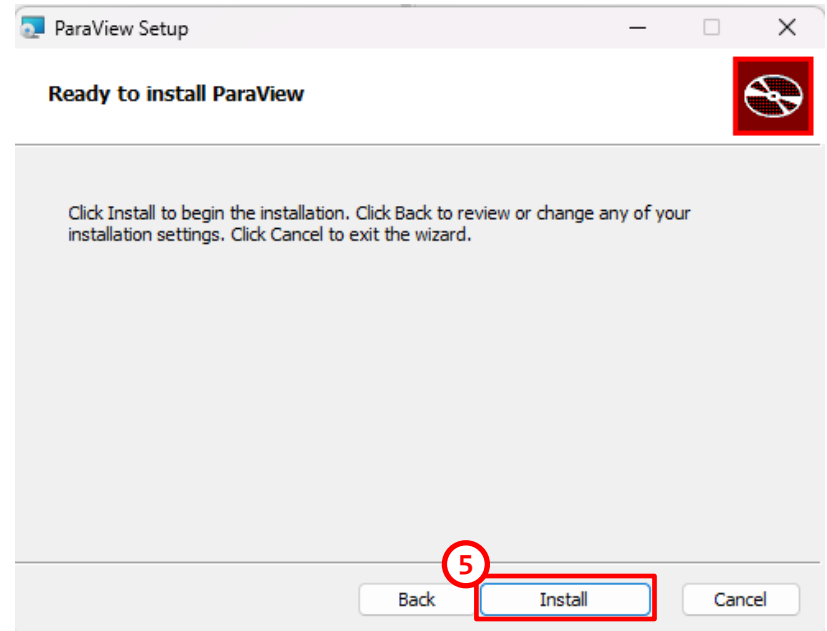
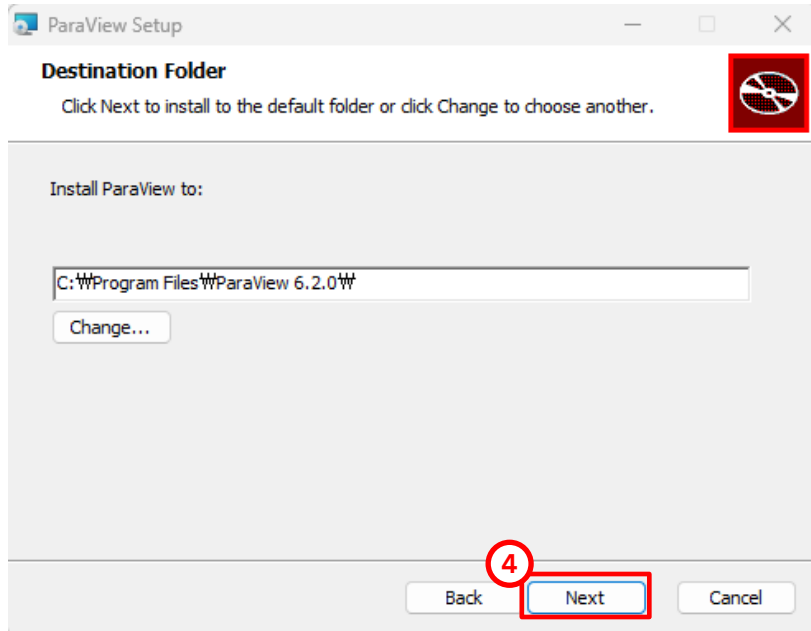
✓ ParaView 설치 파일 실행 후 아래 그림 따라 설치 진행



ParaView 설치 가이드

Step 2. ParaView 설치

- ✓ ParaView 설치 파일 실행 후 아래 그림 따라 설치 진행

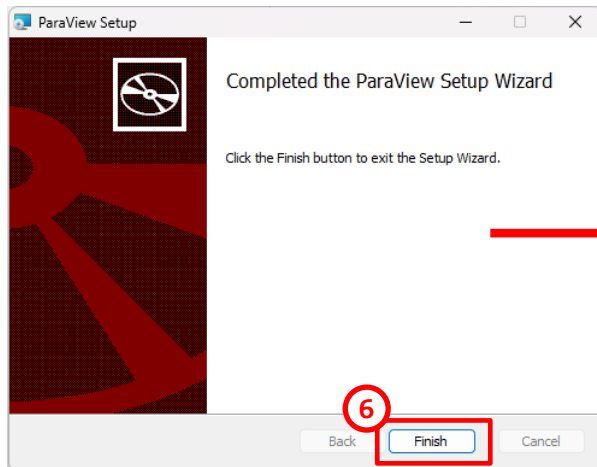


ParaView 설치 가이드

Step 2. ParaView 설치

- ✓ ParaView 설치 파일 실행 후 아래 그림 따라 설치 진행

설치완료



ParaView

